José Carlos Ruiz Sánchez

Full-Stack Developer

Node.js | Express.js | React | Next.js | TypeScript | PostgreSQL | PostGIS | Docker

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SUMMARY

Full-stack developer with a BSc in Computer Engineering and an MSc in Computer Engineering (Big Data) in progress at the University of Almería. During a six-month internship at the Applied Computing Group (ACG), I led most of the development of UAL Trace, a web app that maps campus occupancy from Wi-Fi-connected devices with data refreshed every two minutes. I worked across the PostGIS data model, the n8n ingestion pipeline, the Node.js/Express REST API, the Next.js front end and the Docker deployment. On my own initiative I built Tokimori, a full-stack app with six Node.js/TypeScript microservices behind an nginx API gateway and a React 19 front end.

PORTFOLIO WEBSITE

jrs407.github.io/web-personal

Extended write-ups of every project and role, with technical decisions, architecture, demo videos and downloadable thesis reports, in Spanish and English. Built from scratch in HTML, CSS and JavaScript with no frameworks and deployed on GitHub Pages. Source code at github.com/jrs407/web-personal.

TECHNICAL SKILLS

Languages	JavaScript · TypeScript · Python · SQL
Back end	Node.js · Express.js · Flask · REST APIs · OpenAPI/Swagger · Microservices
Front end	React · Next.js · Vite · React Router · Leaflet · D3.js · Chart.js · HTML5 Canvas · CSS Modules · Tailwind CSS
Databases	PostgreSQL · PostGIS · MySQL
DevOps	Docker · Docker Compose · nginx (reverse proxy and API gateway) · Git
Testing	Vitest · Jest · Supertest · React Testing Library
Security	JWT · bcrypt · helmet · Rate limiting · CORS
Data and automation	n8n · Spark
Academic coursework	Java · Spring · JUnit · Angular · R · MongoDB · Cassandra · Power BI

EXPERIENCE

Full-Stack Developer (Extracurricular Internship) — Mar 2026 - Sep 2026

Applied Computing Group (ACG), University of Almería | Almería, Spain | 25 h/week, on-site and remote

- Took over a Next.js prototype running on mock data and led most of the remaining development of UAL Trace, coordinating daily with the group lead through feature branches and code reviews.
- Designed the PostgreSQL/PostGIS data model (12 tables, spatial and time-window indexes, versioned migrations) and an n8n pipeline ingesting ~3,000 rows every 2 minutes from the Cisco Meraki location API, with device identifiers pseudonymised using HMAC-SHA512.
- Built a REST API in Node.js, Express.js and TypeScript (~30 operations, documented with OpenAPI/Swagger) covered by more than 200 tests with Vitest and Supertest.
- Wrote about 80% of the current Next.js front end, including Leaflet and D3.js map layers (hexagon grid, heatmap, building occupancy, routes between buildings, floor-level detail), a historical mode with time playback, statistics and PDF reports, covered by component tests with React Testing Library.
- Optimised performance under real load with an in-memory cache that merges identical concurrent requests, the Node.js cluster module, covering indexes and a single endpoint that replaced up to 13 sequential requests.
- Deployed with Docker multi-stage builds behind an nginx reverse proxy.

PROJECTS

Tokimori
Feb 2026 - Aug 2026

Personal project | Full-stack web application | Demo video on YouTube | Source code available on request

- Six microservices (auth, items, collection, notes, objectives/canvas, sessions) in Node.js, Express 5 and TypeScript over a shared 8-table MySQL database, behind an nginx API gateway as the only public entry point.
- React 19 front end with Vite, React Router and TypeScript, including a visual editor built on the HTML5 Canvas API with no external libraries that supports shapes, images, text, free drawing, rotation, layers and undo/redo.
- JWT authentication over secure cookies with bcrypt, helmet, rate limiting and configurable CORS.
- Tested with Vitest and Testing Library on the front end and Jest and Supertest on the back end.
- Containerised with Docker Compose, with separate production and development configurations.

Auscultify
Dec 2024 - Jan 2026

Bachelor's thesis | Graded 9.4/10 | Defended Feb 2026 | github.com/jrs407/Auscultify

- Web app for medical students to practise auscultation with real heart, lung and abdominal recordings in 10-question sessions, with personal statistics, daily streaks and comparison with followed users.
- Nine microservices, eight in Node.js, Express and TypeScript and one in Python and Flask for the recommendation engine, with MySQL 8 and Docker Compose.
- Recommendation system with 11 algorithms, from cold-start baselines to user-based and item-based collaborative filtering with Pearson correlation.
- Full analysis and design (27 functional requirements, use cases, Figma prototypes), a React 19 and TypeScript front end with Chart.js, and two technical reports written in LaTeX.

PuppyClicker
Apr 2025 - Jun 2025

Personal project | Full-stack browser game | github.com/jrs407/PuppyClicker

- Clicker game built with React 19, TypeScript, Node.js, Express 5 and MySQL 8, with a 10-endpoint REST API, a 7-table relational schema and Docker Compose deployment.
- Server-side game logic with JWT authentication, bcrypt and purchases in transactions with row-level locking (SELECT FOR UPDATE) to prevent double spending.

Burnout Predictor
May 2026

Machine learning | Python and Spark | github.com/jrs407/ProyectoIABurnout

- Python model that predicts burnout risk from a set of input criteria, deployed across a Spark node cluster to speed up data processing.

EDUCATION

MSc in Computer Engineering
Oct 2025 - Present

University of Almería

Big Data specialisation covering large-scale databases, distributed processing and data pipelines. Fully online and compatible with full-time employment.

BSc in Computer Engineering
Sep 2021 - Feb 2026

University of Almería

Double specialisation in Software Engineering and Information Systems. Coursework in information security (ISO/IEC 27001), quality management (ISO 9001), risk analysis and GDPR, including a consulting project for a real company.

LANGUAGES

Spanish Native | **English** B2, Cambridge B2 First (FCE)

STRENGTHS

Teamwork | Analytical thinking | Time management | Continuous learning

AVAILABILITY

Immediate. On-site, hybrid or remote, and open to relocating anywhere in Spain.